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Appendix C: Selection of hearing protection devices and de-rating schemes

Methods for selecting hearing protection

The following elaborates on the four methods (<https://www.ontario.ca/document/guide-noise-regulation-under-occupational-health-and-safety-act/regulation#section-8>) briefly mentioned in this Guideline. These methods for selecting Hearing Protection Devices (HPDs) are also set out in section 9.3 of CSA Standard, Z94.2-14, "Hearing Protection Devices- Performance, selection, care and use".

1. Single number reporting method

This method provides a value of attenuation for a (HPD) in decibels. The Noise Reduction Rating (NRR) is a single-number descriptor computed from laboratory measurements involving experienced users. The SNR (SF84) is a single-number descriptor computed from laboratory measurements involving inexperienced users.

Noise reduction rating or NRR method

The NRR of a HPD is computed from laboratory measurements involving optimum-subject fit data based on testing according to ANSI Standard S3.19 (1974), "Method for the Measurement of Real-Ear Protection of Hearing Protectors and Physical Attenuation of Earmuffs", which tests subjects under "laboratory conditions." This method is commonly used by manufacturers to report attenuation on HPDs.

Single number rating (subject fit 84th percentile) or SNR (SF84) method

SNR (SF84) is a value that theoretically should be achieved in a workplace with a well-run hearing loss prevention program by approximately 84% of the user population. It is based on testing according to ANSI/ASA Standard 12.6 (2008), "Methods for Measuring the Real-Ear Attenuation of Hearing Protectors", Method B, hearing protection attenuation test. The Method B data is considered to be more representative of "actual field performance values" for "groups of inexperienced users". The SNR (SF84) results are more accurate and approximate "real world" attenuation results.

2. CSA class method

This method involves the assignment of HPD classes based on octave-band attenuation values measured according to ANSI S3.19. CSA Standard Z94.2-14, "Hearing protection devices - Performance, selection, care and use", pre-assigns HPDs into classes according to defined attenuation ranges. A listing of classes (A/AL, B/BL, or C) assigned to HPDs based on the octave band attenuation values measured according to ANSI Standard S 3.19 are provided in Table 4 of CSA Standard Z94.2-14. Dual protection (e.g. a combination of ear muffs and ear plugs) is recommended for noise exposure exceeding $L_{ex, 8}$ of 105 dBA.

3. Octave band computation method

OB method

This is most complicated method of selecting HPDs and provides the maximum accuracy as long as the noise levels are representative of the entire worker shift noise and assuming individual fit is identical to the fit obtained during the testing.

It requires the measurement of the un-weighted workplace sound levels in the 125, 250, 500, 1000, 2000, 4000, and 8000 Hz octave bands and the octave band attenuation data for the HPD being assessed. A full description of the method is described in clause 9.6.6 and in Appendix B of CSA Standard Z94.2-14, "Hearing protection devices - Performance, selection, care and use".

Below is an example of how to estimate the effective noise level by this method when wearing a particular HPD.

Selection of Hearing Protection Devices and De-Rating Schemes								
Octave-band Centre frequency (Hz)	125	250	500	1000	2000	4000	8000	Overall level
Workplace noise spectrum (octave band sound pressure level)	85	88	93	90	89	87	86	98
A-weighting correction	-16.1	-8.6	-3.2	0	+1.2	+1.0	-1.1	n/a
Unprotected A weighted sound level (octave band sound pressure level minus A-weighting correction)	68.9	79.4	89.8	90	90.2	86	84.9	96

Mean attenuation of HPD (from manufacturer's data)	12	15	20	26	31	37	35	n/a
Standard deviation of attenuation	2.8	3	4	3.7	4.9	5.9	4.0	n/a
Assumed protection values (mean attenuation minus standard deviation of attenuation)	9.2	12	16	22.3	26.1	31.1	31.0	n/a
Effective A-weighted (dBA) sound level when HPD is worn	59.7	67.4	73.8	67.7	64.1	56.9	53.9	76.0

The estimated protected exposure level is calculated using the following equation:

$$\text{Overall protected exposure level } L_{eq} = 10 \log (10^{0.1SPL_1} + \dots + 10^{0.1SPL_n})$$

in which

L_{eq} is the equivalent sound pressure level in dBA

i is a discrete activity of a worker exposed to a sound level

n is the total number of discrete activities in the worker's shift.

SPL_i is the L_{eq} for the ith activity in dBA

$$\text{Overall protected exposure level } L_{eq} = 10 \log (10^{5.97} + 10^{6.74} + \dots + 10^{5.39}) = 76.0 \text{ dBA}$$

CSA Standard Z94.2-14, "Hearing protection devices - Performance, selection, care and use" (i.e. Clause 9.6.6.2) requires use of this method for exposures above an L_{ex} of 105 dBA.

4. Field attenuation estimation system (FAES)

This method involves individual users and implements fit-testing using field attenuation estimation systems (FAES). Unlike the above three methods (where group attenuation is obtained), this objective method helps to determine personal attenuation ratings (PAR). The trend is to use both the individual (fit-testing) and group attenuation approaches to obtain a more conservative estimate of a hearing protection device's attenuation and worker's protected exposure.

De-rating of HPDs

Much of the sound attenuation data in use today is still from ANSI S3.19-1974, "Method for the Measurement of Real-Ear Protection of Hearing Protectors and Physical Attenuation of Earmuffs", including the NRRs (Noise Reduction Ratings), that manufacturers print on their packages of HPDs.

Noise reduction rating (NRR) and de-rating

Since the NRR is a laboratory test involving experienced users and may not reflect real-world results, HPDs must be "de-rated" to account for the significantly reduced protection provided under "real world" conditions. CSA Standard Z94.2-14, "Hearing protection devices - Performance, selection, care and use", recommends de-rating NRRs by the following multiplicative factors:

- 70% (0.7) for earmuffs
- 50% (0.5) for earplugs
- 65% (0.65) for dual protection

Use with A- weighted sound measurements

The NRR method was designed for use with C-weighted data as follows:

Predicted A-weighted exposure = C-weighted exposure - NRR

To determine the exposure in dBA, CSA Standard Z94.2-14, Hearing Protection Devices- Performance, selection, care and use, suggests that the NRR may be applied to A-weighted sound level data in place of C-weighted data, by adding a +3 dB correction factor to the A-weighted sound level to estimate the C-weighted sound level as set out in the formula below.^[1]

Predicted A-weighted exposure = A-weighted exposure + 3dB - NRR

Taking into account both, the de-rating of NRRs for the various types of HPDs, and the adjusting for the difference between C-weighted and A-weighted sound level measurements, a determination may be made of the predicted A-weighted exposure when the HPD is worn.

This approach is reflected in the scheme set out in Table 2 of CSA Standard Z94.2-14, "Hearing Protection Devices- Performance, selection, care and use".

Noise Reduction Rating (NRR) and De-Rating		
Device Type	% of NRR Achieved	Predicted dBA effective at the ear
Earplugs	50%	$L_{eq} - [NRR (0.5) - 3]$
Earmuffs	70%	$L_{eq} - [NRR (0.7) - 3]$
Dual Protection	65%	$L_{eq} - [(NRR+5)(0.65) - 3]$

The following example illustrates the use with A-weighted sound measurements. If the user wishes to apply

the NRR to an A-weighted sound level, then the NRR is to be reduced by 3 dB after the de-rating, based on the type of HPD, is applied.

For a measured A-weighted L_{eq} of 94 dBA, and earplugs (with a labelled NRR of 28), the predicted A-weighted effective L_{eq} is calculated as follows:

$$L_{eq} - [NRR (0.5) - 3] = XX \text{ dBA}$$

$$94 \text{ dBA} - [28 (0.5) - 3] = 94 - 11 = 83 \text{ dBA}$$

If the user wishes to apply the NRR to a C-weighted sound level, then the NRR remains the same after the de-rating is applied. For a measured C-weighted L_{eq} of 94 dBC, and earplugs (with a labelled NRR of 28), the predicted A-weighted effective L_{eq} is calculated as follows:

$$94 \text{ dBC} - [28 (0.5)] = 94 - 14 = 80 \text{ dBA}$$

[1] CSA Standard Z94.2-14, "Hearing Protection Devices – Performance, selection, care and use", cites the works of Gauger and Berger (2004) as the basis for 3 dB correction factor.

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